

Expt 7	Spectral Estimation
Date:	7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

Code:

```

clc; clear; close all
fs=1000; t=(0:999)/fs;
s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);
n=0.3*randn(size(t));
x=s+n;
figure, plot(x), grid on title('Signal with Noise')
figure, periodogram(x,[],[],fs)
title('PSD using Periodogram')
figure
X=abs(fft(x));
f=(0:length(X)-1)*fs/length(X);
plot(f(1:500),X(1:500)), grid on
title('Magnitude Spectrum')
figure
bar([mean(s.^2) var(n)])
set(gca,'XTickLabel',{'Signal','Noise'})
title('Power Comparison')
grid on

```

Outputs:





